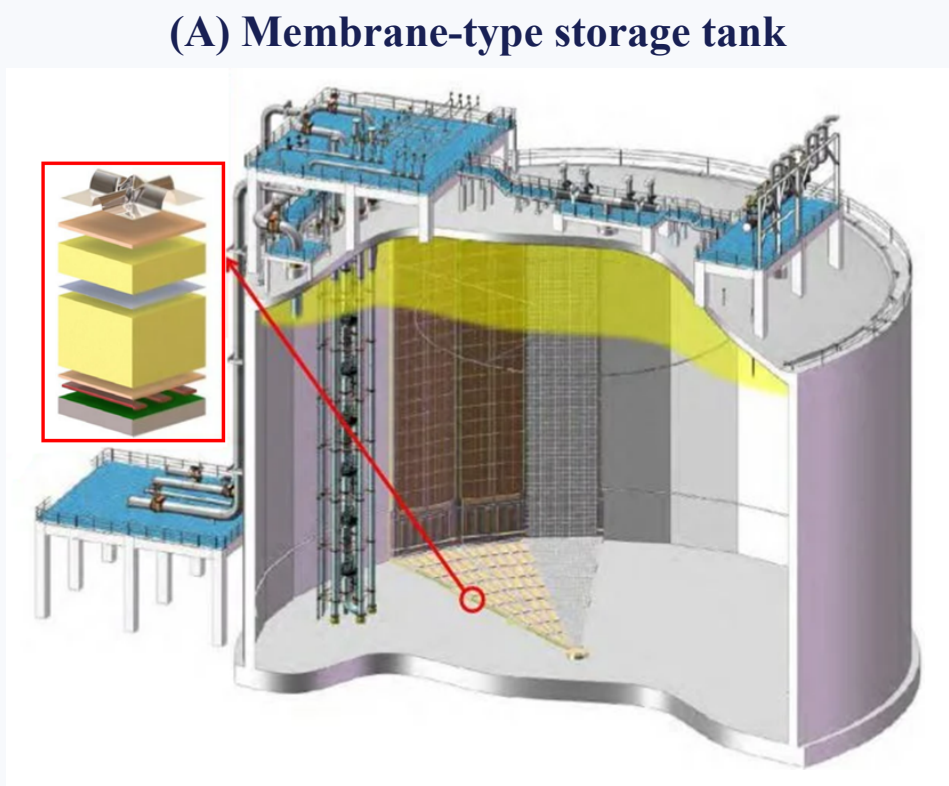




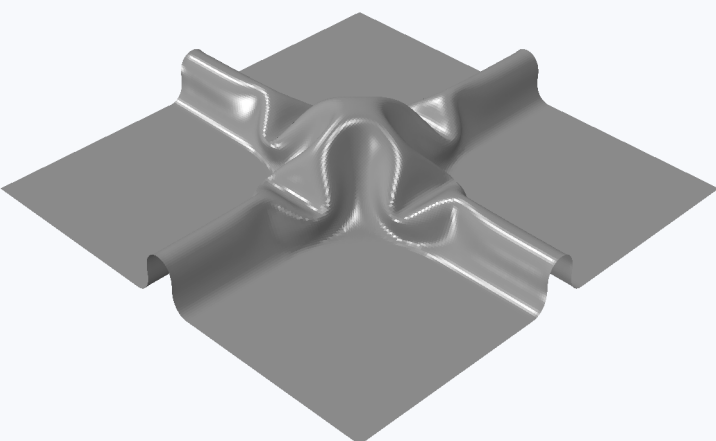
Introduction

Large-scale liquid hydrogen (LH₂) cargo containment requires thin metallic barriers that remain gas-tight while accommodating thermal contraction and restrained in-plane deformation. In our previous work, an orthogonally corrugated 316L stainless-steel membrane was proposed and numerically optimized for LH₂ thermal-contraction and pressure-difference load cases, showing that corrugation can convert global contraction into local bending deformation and reduce strain demand in planar regions.

The present work experimentally validates this deformation-accommodation mechanism using Digital Image Correlation (DIC) under tensile loading, Acoustic Emission (AE) monitoring during cyclic testing, and liquid-nitrogen (LN₂) thermal cycling of a 0.2 m³ demonstrator. Mechanical testing at LH₂ temperature is deferred because hydrogen-rated cryogenic testing requires dedicated safety infrastructure; room-temperature tests isolate the geometric mechanism, while LN₂ cycling provides a non-flammable cryogenic integrity screen.



(B) Corrugated membrane geometry



(C) Experimental validation route

